

RESEARCH/WORK EXPERIENCE

INRIA, SODA TEAM

Machine Learning Research Assistant

September 2025 - Present
(full-time through January 2026, part-time thereafter)
Palaiseau, France

- **First author "Pan-Cancer versus Single-Cancer Training for Survival Prediction Using Large-Scale TCGA and SEER Clinical Cohorts", in preparation for submission to GigaScience, with Gaël Varoquaux and Judith Abécassis (INRIA SODA)**
- Quantifying the efficacy of pan-cancer vs. single-cancer training by harmonizing the TCGA (11k patients) and SEER (7M patients) cohorts, creating one of the largest unified clinical datasets for survival prediction
- Utilized advanced tabular representation learning (SurvivalBoost, TARTE, sentence embeddings) to evaluate model transferability across 31 shared clinical features
- Developed a robust data pipeline to resolve cross-cohort discrepancies, including sentinel normalization, TNM derivation, and patient-level deduplication

BAR-ILAN UNIVERSITY

Machine Learning Research Assistant

June 2026 - Present
Ramat Gan, Israel

- Worked with Prof. Ofir Lindenbaum on semi-supervised learning methods for tabular data
- Developed and evaluated approaches for leveraging unlabeled data in tabular prediction tasks

EPFL (École Polytechnique Fédérale de Lausanne)

Machine Learning Research Assistant

June 2024 - September 2025
(full-time through September 2024, part-time thereafter)
Lausanne, Switzerland

- Worked under the supervision of Professor Pascal Fua, Romuald Houdré and Nicolas Villa in the Laboratory of Quantum and Nano-Optics (LQNO)
- Pioneered the application of time-series and tabular ML classifiers (XGBoost, MultiRocket, HIVE-COTE 2.0) for antimicrobial susceptibility testing, representing the first-ever use of these architectures with photonic crystal optical transmission data
- Engineered a custom pipeline to translate raw optical signals into high-accuracy biological sensing predictions : https://github.com/dansuissa/EPFL_AMR

INSERM

Machine Learning Research Assistant (Bachelor Thesis)

January - March 2025
Paris, France

- Worked under the supervision of Professor Guillaume Fillion, Ariel Lindner and Amir Pandi
- Extended a codon language model, CodonTransformer (nature.com/articles/), to optimize regulatory DNA sequences using novel tokenization and embedding techniques, enhancing biological sequence encoding : <https://github.com/dansuissa/CodonTransformerV2>

ÉCOLE POLYTECHNIQUE

Undergraduate Machine Learning Research Assistant

September - December 2024
Palaiseau, France

- In collaboration with Pr. Davide Buscaldi, evaluated LLM unlearning methods on the SemEval-2025 dataset to reduce memorization of sensitive data beyond output-filtering baselines

EDUCATION

MCGILL UNIVERSITY

MEng Applied Artificial Intelligence

January 2026 - May 2027 (expected dates)
Montreal, Canada

ÉCOLE POLYTECHNIQUE

BSc Mathematics and Computer Science

August 2022 - September 2025
Palaiseau, France

STANFORD UNIVERSITY

International Honors Program

June - August 2023
Stanford CA, USA

Relevant courses: Theory of Machine Learning (MATH 562), Network Science (COMP 511), Optimization and Optimal Control (ECSE 507), Machine Learning for Engineers (ECSE 551), Stochastic Processes, Compilers, Design and Analysis of Algorithms, Computer Architecture, Complexity Analysis

PROJECTS

- **Theory of Machine Learning Project:** Combined proof-based machine learning theory with empirical analysis to derive and validate scaling laws for overparameterized linear regression and two-layer neural networks, with a focus on SGD dynamics, generalization, and NTK, mean-field, and random-features limits

SKILLS

Python (TensorFlow, scikit-learn, skrub, PyTorch, numpy, pandas), R, C/C++, Git, Linux, Coq

LANGUAGES

- French (Native)
- English (Fluent)
- Spanish (Intermediate)
- Hebrew (Intermediate)

VOLUNTEERING & EXTRACURRICULARS

Scout movement group leader · Maths Olympiads tutor · IANTD Rescue Diver · MMA assistant coach